

JOATHAM JACOB

Energy and Decarbonisation Engineer
Industrial energy efficiency · Marginal abatement cost modelling · Energy management systems (ISO 50001)
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Italian residence permit, work-authorised in Italy · Open to relocation

PROFESSIONAL EXPERIENCE

ESG Supply Chain & International Project Sourcing Intern — Industrious Global Technologies, Milan, Italy **Feb 2025 – Aug 2025**

Built the company's 2024 Scope 1 and 2 emissions inventory from primary consumption data under the GHG Protocol, setting the boundary and emission factors and running formal quality controls on internal and supplier data. Screened international suppliers against ESG criteria, flagged efficiency opportunities across operations, and built the KPI reporting used by management.

ESG Measurement Intern — Omtra S.r.l. Società Benefit, Milan, Italy **Nov 2023 – May 2024**

Carried out a cradle-to-grave product LCA for Cartier and calculated Scope 1–3 emissions for client companies from data collected on site. For Tuscan textile maker Busatti, collected energy and operational data across two production sites, built its carbon footprint, contributed to its first statutory Impact Report and supported its B Impact Assessment. Supported B Corp and EcoVadis verification across clients by testing reported data against source evidence.

ESG Analyst (Remote) — Earth Academy Global, Amstelveen, Netherlands **May 2021 – Nov 2022**

Analysed corporate sustainability disclosures against five frameworks (GRI, SASB, TCFD, Integrated Reporting, BRSR) for clients across Europe, the Middle East and India. Prepared materials for client trainings delivered with ISOS Group, including a GRI training in Dubai and an ESG masterclass for Al-Dabbagh Group (Saudi Arabia), and supported training for sustainability professionals at Tanmiah Food Company. Joined as an intern building an e-learning course on India's new BRSR framework, then moved into the full-time analyst role.

TECHNICAL PROJECTS AND FIELD WORK

Industrial decarbonisation assessment, Acciaieria Arvedi, Cremona (2026)

Rebuilt the Scope 1 baseline at 434,844 tCO_2 (EU ETS, CY2023) and showed that its net-zero claim rests on three accounting instruments and one physical reduction.

- Costed a 2030 MACC: 192,494 tCO_2/yr (44.3%) at a modelled €58.0M capex and €11.44 per tonne of steel. Capex and opex are engineering estimates, not vendor quotes.
- Found that 59% of the abatement depends on ~36 kt/yr of certified biochar, and that ETS and CBAM costs rise from €6.9M to €32.6M/yr.

Energy monitoring pipeline (Python, Power BI, 2026): degree-day regression baseline with deviation flagging for ISO 50001-style reporting.

Web tools (jacob-hub-esg.github.io, 2026): four live tools — a MACC and project-finance calculator, the BuildingCarbon Scope 1–2 benchmark, a CSRD readiness assessment (10 ESRS) and a CO_2 calculator.

Field audits (B.Tech): energy, water and safety audits of 8 railway stations in Bihar over 25 days, covering load analysis, consumption measurement and site-specific reports with prioritised interventions. Also evaluated a biogas plant's performance in Palakkad.

EDUCATION

M.Sc. Environmental Change and Global Sustainability — Università degli Studi di Milano, Italy **2022–2025**

- Classe LM-75. Full academic scholarship. Final grade 100/110. Taught entirely in English.
- **Thesis**: Emission Reduction Commitments in Global Agriculture: A Comparative Analysis of Corporate Climate Disclosures.

B.Tech Energy and Environmental Engineering — Tamil Nadu Agricultural University, India **2016–2021**

- **Thesis:** Design and Fabrication of a Solar Powered Trash Compactor (published, TNAU 2020).
- **Coursework**: energy auditing, renewable energy systems, thermodynamics, environmental impact assessment.

SKILLS

- Energy management: ISO 50001 (energy review, SEUs, EnPIs, energy baselines) · energy auditing · MACC · M&V principles · industrial energy systems (motors and VFDs, compressed air, steam, process heat) · heat electrification · waste-heat recovery
- Analysis and finance: NPV · IRR · LCOE · levelised cost of abatement · payback · scenario and sensitivity analysis · regression baselining
- Carbon and regulation: GHG Protocol Scope 1–3 · ISO 14064-1/-2 · LCA · EU ETS · CBAM · CSRD/ESRS · EU Taxonomy · SBTi · GRI · SASB · TCFD · B Corp · EcoVadis
- Familiar with: ISO 14001 · ISO 19011 · IPMVP · EED recast and D.Lgs. 102/2014 energy audits · Italian energy bills (POD/PDR, F1–F3 bands) · Conto Termico 3.0 · Certificati Bianchi
- Tools: Excel (advanced: Power Query, pivots, modelling) · Power BI · Python (pandas) · React · Tableau · PV Syst · AutoCAD · QGIS
- Languages: English (C1, IELTS 7.5) · Italian (B1) · Malayalam (native) · Tamil · Hindi

CERTIFICATIONS

- GHG 202: Basics of Project-Level GHG Accounting — GHG Management Institute (in progress, 2026)
- GHG Accounting: Corporate Standard — GHG Protocol (2026)
- ISO 50001 Energy Management Systems — Coursera (2026); prepared to pursue the ISO 50001 Lead Auditor pathway
- Energy Audits and Energy Efficiency Fundamentals — Schneider Electric University (2026)
- IGBC Accredited Professional
- Earth Academy Practitioner: GRI, TCFD, Integrated Reporting, SDGs